used as a source of energy in rural areas. In bio-remediation naturally occurring or genetically engineered microorganisms are used to reduce or degrade pollutants.

Evaluation

1. Which of the following microorganism is used for production of citric acid in industries?

   a) *Lactobacillus bulgaris*
   b) *Penicillium citrinum*
   c) Aspergillus niger
   d) *Rhizopus nigricans*

2. Which of the following pair is correctly matched for the product produced by them?

   a) *Acetobacter aceti* - Antibiotics
   b) *Methanobacterium* - Lactic acid
   c) *Penicilium notatum* - Acetic acid
   ***d) Saccharomyces cerevisiae*- Ethanol**

3. The most common substrate used in distilleries for the production of ethanol is_____

   a) Soyameal b) Groundgram
   c) Molasses d) Corn meal

4. Cyclosporin – A is an immunosuppressive drug produced from _____

   a) *Aspergillus niger*
   b) *Manascus purpureus*
   c) *Penicillium notatum*
   d) Trichoderma polysporum

5. CO_2 is not released during

   a) Alcoholic fermentation
   b) Lactate fermentation
   c) Aerobic respiration in animals
   d) Aerobic respiration in plants

6. The purpose of biological treatment of waste water is to _____

   a) Reduce BOD b) Increase BOD
   c) Reduce sedimentation
   d) Increase sedimentation

7. The gases produced in anaerobic sludge digesters are

   a) Methane, oxygen and hydrogen sulphide.
   b) Hydrogen sulphide, methane and sulphur dioxide.
   c) Hydrogen sulphide, nitrogen and methane.
   d) Methane, hydrogen sulphide and CO_2.

8. How is milk converted into curd? Explain the process of curd formation.

9. Give any two bioactive molecules produced by microbes and state their uses.

10. Define the following terms:

    a) Antibiotics b) Zymology
    c) Superbug

11. Write short notes on the following.

    a) Brewer's yeast b) *Ideonella sakaiensis*
    c) Microbial fuel cells

12. List the advantages of biogas plants in rural areas.

13. When does antibiotic resistance develop?

14. What is referred to as industrial alcohol? Briefly describe its preparation.

15. What is bioremediation?

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